Analysis of Digital Payment Adoption in Tunisia

COURSE:

BUSINESS INTELLIGENCE AND
DATABASE MANAGEMENT SYSTEMS

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1. Introduction

This BI report analyzes **digital payment adoption in Tunisia** via Flouci, a local digital wallet platform. The project aims to provide **data-driven insights** into user behavior, transaction patterns, and wallet performance to guide marketing, product optimization, and retention strategies.

Key focus areas:

- **Customer Behavior**: Age, gender, income, education, wallet type, tenure, and attrition
- **Transaction Patterns**: Amounts, frequency, wallet tiers, and regional distribution
- **Strategic Decisions**: Insights to improve adoption and retention

Objectives:

- Analyze adoption trends across customer segments
- Identify high-risk customers with potential attrition
- Evaluate wallet tier performance
- Provide actionable recommendations for Flouci management

2. Implementation

2.1 Data Gathering

2.1.1 Data Collection

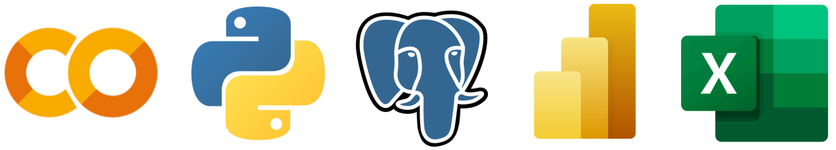
Data sourced from anonymized Flouci transactions (~10,000 records), including:
customer_id, age, gender, income_category, education_level, wallet_category, customer_tenure_months, transaction_amount, transaction_count, attrition_flag
The dataset was adapted to the Tunisian context using World Bank financial inclusion data.

2.1.2 Initial Data Review :

- Explored dataset structure and data types
- Identified missing values and outliers
- Checked categorical distributions (gender, wallet type, income, education)

Drive / GitHub Link: (Insert here)

Software Used:



2.2 Data Preparation

2.1 Data Gathering

The raw dataset was initially imported in its original format in order to preserve the integrity of the data before any transformation or preprocessing.

id	action_type	action_tim	device_typ	os_type	location_c	location_st	location_c	transactio	transactio	transactio	age	gender	education_	income_br	employme	has_bank_	is_first_tim	barrier_rea	driver_reas	registra
000	registratio	2024-05-0	mobile	iOS	New York	NY	USA				29	female	master	high	employed	TRUE	TRUE		convenien	2024-05
000	payment	2024-05-0	mobile	Android	London	ENG	UK	54,25	GBP	success	32	male	bachelor	medium	employed	TRUE	FALSE		rewards	2023-12
000	add_card	2024-05-0	mobile	Android	Delhi	DL	India				22	male	high_scho	low	student	FALSE	TRUE			2024-05
000	withdrawa	2024-05-1	desktop	Windows	Paris	IDF	France	1100	EUR	success	52	female	doctorate	high	employed	TRUE	FALSE		merchant_	2021-11
000	failed_atte	2024-05-1	tablet	Other	Sydney	NSW	Australia				48	female	secondary	medium	unemploye	FALSE	FALSE	usability_is		2022-08
000	registratio	2024-05-1	mobile	Android	Mumbai	MH	India				19	male	high_scho	low	student	FALSE	TRUE		peer_influe	2024-05
000	login	2024-05-1	desktop	Windows	Berlin	BE	Germany				41	female	bachelor	medium	employed	TRUE	FALSE			2020-03
000	payment	2024-05-2	mobile	iOS	Chicago	IL	USA	250,75	USD	success	27	female	bachelor	medium	employed	TRUE	FALSE		convenien	2022-05
000	add_card	2024-05-2	mobile	Android	Bangkok	BKK	Thailand				24	female	high_scho	low	student	FALSE	TRUE			2024-05
000	withdrawa	2024-05-2	desktop	MacOS	Toronto	ON	Canada	760,5	CAD	success	65	male	doctorate	high	retired	TRUE	FALSE		security	2017-03
000	update_pr	2024-05-2	mobile	iOS	New York	NY	USA				37	male	master	high	employed	TRUE	FALSE			2021-10
000	failed_atte	2024-05-2	tablet	Other	Johannesb	GT	South Afric				23	male	secondary	low	unemploye	FALSE	TRUE	no_interne		2024-05
000	payment	2024-05-2	mobile	Android	Madrid	MD	Spain	89,99	EUR	success	36	female	bachelor	medium	employed	TRUE	FALSE		convenien	2021-08
000	registratio	2024-05-2	desktop	Windows	Boston	MA	USA				47	male	doctorate	high	employed	TRUE	TRUE		security	2024-05
000	withdrawa	2024-05-3	mobile	iOS	Tokyo	TKY	Japan	500	JPY	success	34	female	master	medium	employed	TRUE	FALSE		merchant_	2022-02
000	failed_atte	2024-05-3	tablet	Linux	Lima	LMA	Peru				21	other	secondary	low	student	FALSE	TRUE	security_c		2024-05
000	view_offer	2024-06-0	desktop	MacOS	San Franci	CA	USA				40	female	bachelor	medium	employed	TRUE	FALSE			2020-10
000	payment	2024-06-0	mobile	Android	Dubai	DU	UAE	1250	AED	success	38	male	master	high	employed	TRUE	FALSE		rewards	2019-06
000	logout	2024-06-0	mobile	Android	Mexico Cit	CDMX	Mexico				43	female	bachelor	medium	employed	TRUE	FALSE			2018-11
000	registratio	2024-06-0	mobile	Android	Moscow	MOW	Russia				31	male	bachelor	medium	employed	TRUE	TRUE		convenien	2024-06
000	failed_atte	2024-06-0	tablet	Other	Istanbul	IST	Turkey				20	female	high_scho	low	student	FALSE	TRUE	lack_of_tru		2024-06
000	add_card	2024-06-0	desktop	Windows	Seattle	WA	USA				34	male	bachelor	medium	employed	TRUE	FALSE			2022-04
000	payment	2024-06-0	mobile	Android	Brussels	BRU	Belgium	72,5	EUR	success	45	male	master	high	employed	TRUE	FALSE		convenien	2015-05
000	withdrawa	2024-06-1	desktop	Windows	Jakarta	JKT	Indonesia	300	IDR	success	39	female	bachelor	medium	employed	TRUE	FALSE		merchant_	2014-04
000	update_pr	2024-06-1	mobile	iOS	Zurich	ZH	Switzerlan				50	female	doctorate	high	employed	TRUE	FALSE			2008-06
000	registratio	2024-06-1	mobile	Android	Hong Kong	HK	Hong Kong				26	female	bachelor	medium	employed	TRUE	TRUE		convenien	2024-06
000	payment	2024-06-1	desktop	Windows	Athens	AT	Greece	230	EUR	success	28	male	bachelor	medium	employed	TRUE	FALSE		convenien	2022-03
000	view_offer	2024-06-1	desktop	MacOS	Cape Towr	WC	South Afric				33	female	master	medium	employed	TRUE	FALSE			2019-04
000	withdrawa	2024-06-1	desktop	Windows	Oslo	OS	Norway	900	NOK	success	59	male	doctorate	high	retired	TRUE	FALSE		security	2010-11
000	failed_atte	2024-06-2	tablet	Other	Bogota	BOG	Colombia				17	male	secondary	low	student	FALSE	TRUE	no_smartp		2024-06

2.2 Data Preparation

The dataset was then cleaned and standardized using Microsoft Excel, where irrelevant columns were removed, categorical values were unified, and data quality checks were performed to ensure consistency and reliability.

	A	B	C	D	E	F	G	H	I	J	K	L
1	customer_i	activity_statu	age	gend	dependents	education_lev	marital_statu	income_catego	card_catego	customer_tenure_mont	transaction_amou	transaction_cou
2	768805383	Existing Customer	45	M	3	High School	Married	$60K - $80K	Blue	39	1144	42
3	818770008	Existing Customer	49	F	5	Graduate	Single	Less than $40K	Blue	44	1291	33
4	713982108	Existing Customer	51	M	3	Graduate	Married	$80K - $120K	Blue	36	1887	20
5	769911858	Existing Customer	40	F	4	High School	Unknown	Less than $40K	Blue	34	1171	20
6	709106358	Existing Customer	40	M	3	Unknown	Married	$60K - $80K	Blue	21	816	28
7	713061558	Existing Customer	44	M	2	Graduate	Married	$40K - $60K	Blue	36	1088	24
8	810347208	Existing Customer	51	M	4	Unknown	Married	$120K +	Gold	46	1330	31
9	818906208	Existing Customer	32	M	0	High School	Unknown	$60K - $80K	Silver	27	1538	36
10	710930508	Existing Customer	37	M	3	Unknown	Single	$60K - $80K	Blue	36	1350	24
11	719661558	Existing Customer	48	M	2	Graduate	Single	$80K - $120K	Blue	36	1441	32
12	708790833	Existing Customer	42	M	5	Unknown	Unknown	$120K +	Blue	31	1201	42
13	710821833	Existing Customer	65	M	1	Unknown	Married	$40K - $60K	Blue	54	1314	26
14	710599683	Existing Customer	56	M	1	Graduate	Single	$80K - $120K	Blue	36	1539	17
15	816082233	Existing Customer	35	M	3	Graduate	Unknown	$60K - $80K	Blue	30	1311	33
16	712396908	Existing Customer	57	F	2	Graduate	Married	Less than $40K	Blue	48	1570	29
17	714885258	Existing Customer	44	M	4	Unknown	Unknown	$80K - $120K	Blue	37	1348	27
18	709967358	Existing Customer	48	M	4	Graduate	Single	$80K - $120K	Blue	36	1671	27
19	753327333	Existing Customer	41	M	3	Unknown	Married	$80K - $120K	Blue	34	1028	21
20	806160108	Existing Customer	61	M	1	High School	Married	$40K - $60K	Blue	56	1336	30
21	709327383	Existing Customer	45	F	2	Graduate	Married	Unknown	Blue	37	1207	21
22	806165208	Existing Customer	47	M	1	Graduate	Divorced	$60K - $80K	Blue	42	1178	27
23	708508758	Attrited Customer	62	F	0	Graduate	Married	Less than $40K	Blue	49	692	16
24	784725333	Existing Customer	41	M	3	High School	Married	$40K - $60K	Blue	33	931	18
25	811604133	Existing Customer	47	F	4	Unknown	Single	Less than $40K	Blue	36	1126	23
26	789124683	Existing Customer	54	M	2	Unknown	Married	$80K - $120K	Blue	42	1110	21
27	771071958	Existing Customer	41	F	3	Graduate	Single	Less than $40K	Blue	28	1051	22
28	720466383	Existing Customer	59	M	1	High School	Unknown	$40K - $60K	Blue	46	1197	23
29	804424383	Existing Customer	63	M	1	Unknown	Married	$60K - $80K	Blue	56	1904	40
30	718813833	Existing Customer	44	F	3	Unknown	Single	Unknown	Blue	34	1052	18
31	806624208	Existing Customer	47	M	4	High School	Married	$40K - $60K	Blue	42	1045	38
32	778348233	Existing Customer	53	M	3	Unknown	Married	$80K - $120K	Blue	33	1038	25
33	712991808	Existing Customer	53	M	2	Unknown	Married	$60K - $80K	Blue	48	1596	26

2.2 Data Preparation

The data dictionary provides a detailed description of each variable used in the analysis, including its definition, data type, and an example value. It serves as a reference to ensure a clear understanding of the dataset structure and to maintain consistency during data preparation, analysis, and dashboard development.

Column Name	Description	Data Type	Example
customer_id	Unique identifier for each customer	Integer	769000123
activity_status	Indicates wether the customer is active or inactive in digital payments	Text	Active
age	customer age	Integer	45
gender	customer gender	Text	M
dependents	number of dependents	Integer	2
education_level	customer education category	Text	Graduate
marital_status	customer marital category	Text	Married
income_category	customer income range	Text	$60K-$80K
card_category	type of bank card	Text	Gold
customer_tenure_months	number of months as a customer	Integer	36
transaction_amount	total amount of digital payment transactions	Numeric	12450
transaction_count	number of digital payment transactions	Integer	85

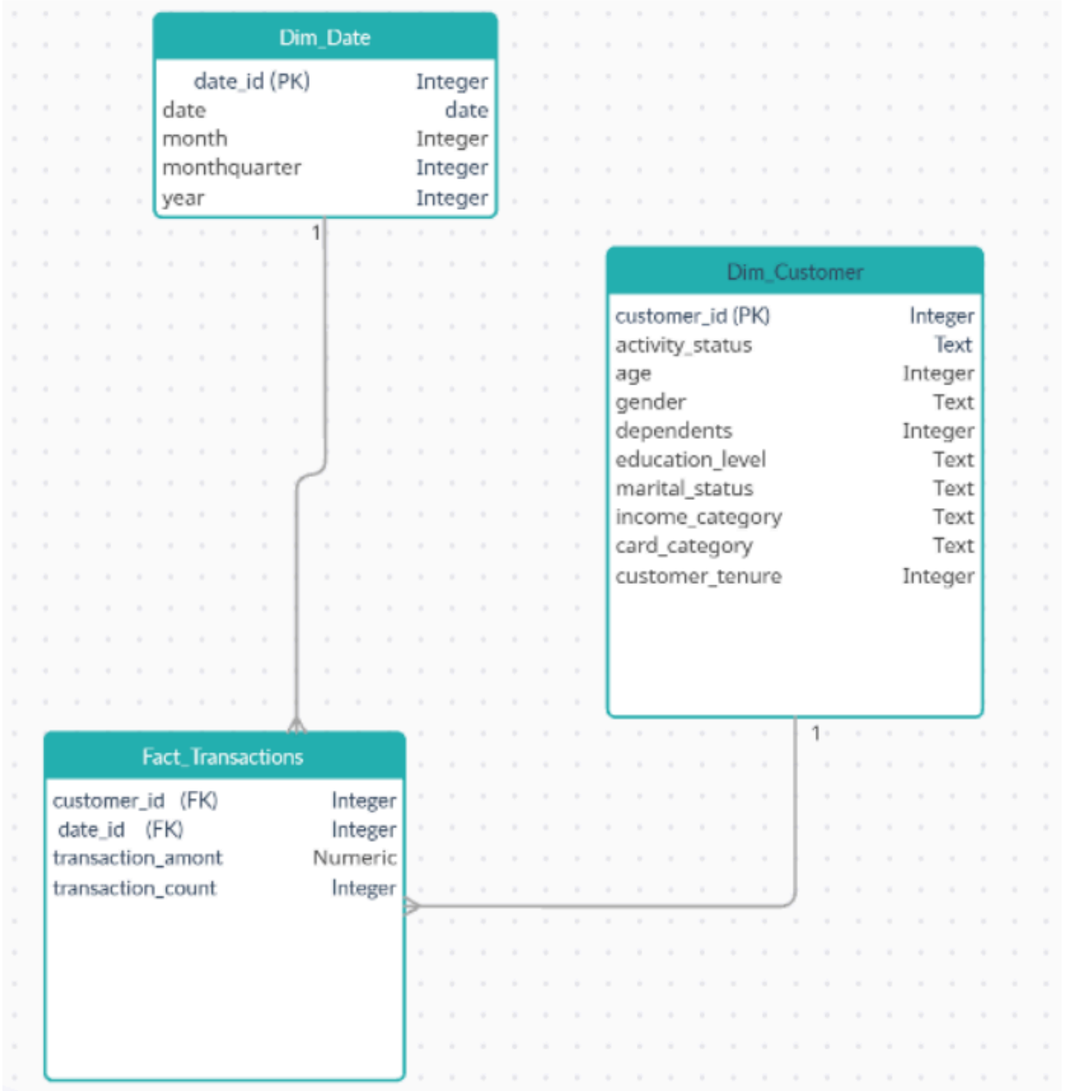
2.2.3 Schema Design

Fact Table:

- fact_transactions → transaction_id, customer_id, date_id, transaction_amount, transaction_count

Dimension Tables:

- dim_customer → customer_id, age, gender, income_category, education_level, marital_status, wallet_category, customer_tenure
- dim_date → date_id, date, month, quarter, year
- dim_product → wallet_category, product_features



1. Creation FACT/DIMENSION tables :

```python
# Create Customer Dimension
customer_dim = bank_df.loc[:, [
    'customer_id',
    'activity_status',
    'age',
    'gender',
    'dependents',
    'education_level',
    'marital_status',
    'income_category',
    'card_category',
    'customer_tenure_months'
]].copy()

# Drop duplicate customers
customer_dim.drop_duplicates(subset='customer_id', inplace=True)

# Reset index
customer_dim.reset_index(drop=True, inplace=True)
```

1. Creation FACT/DIMENSION tables :

```
import pandas as pd

# Convert transaction_date to datetime
bank_df['transaction_date'] = pd.to_datetime(bank_df['transaction_date'])

# Create Date Dimension
date_dim = bank_df[['transaction_date']].drop_duplicates()

# Extract date parts
date_dim['year'] = date_dim['transaction_date'].dt.year
date_dim['quarter'] = date_dim['transaction_date'].dt.quarter
date_dim['month'] = date_dim['transaction_date'].dt.month
date_dim['day'] = date_dim['transaction_date'].dt.day

# Reset index
date_dim.reset_index(drop=True, inplace=True)
```

1. Creation FACT/DIMENSION tables :

```
# Create Fact Transactions table
fact_transactions = bank_df.loc[:, [
    'customer_id',
    'transaction_amount',
    'transaction_count'
]].copy()

# Reset index
fact_transactions.reset_index(drop=True, inplace=True)
```

2.Create dimensions/fact tables in the DB

```sql
-- 1. Create the database
CREATE DATABASE bank_transactions;
```

```sql
-- 3. Create Dim_Date table
CREATE TABLE Dim_Date (
    date_id SERIAL PRIMARY KEY,
    date DATE NOT NULL UNIQUE,                -- each date should be unique
    month INTEGER NOT NULL CHECK (month BETWEEN 1 AND 12),
    monthquarter INTEGER NOT NULL CHECK (monthquarter BETWEEN 1 AND 4),
    year INTEGER NOT NULL CHECK (year > 1900)    -- assuming realistic years
);
```

```sql
-- 4. Create Dim_Customer table
CREATE TABLE Dim_Customer (
    customer_id SERIAL PRIMARY KEY,
    activity_status TEXT NOT NULL CHECK (activity_status IN ('Active','Inactive')),
 -- only valid statuses
    age INTEGER NOT NULL CHECK (age >= 0),        -- no negative ages
    gender TEXT CHECK (gender IN ('Male','Female','Other')),  -- optional
    dependents INTEGER DEFAULT 0 CHECK (dependents >= 0),
    education_level TEXT,
    marital_status TEXT CHECK (marital_status IN
('Single','Married','Divorced','Widowed')),
    income_category TEXT,
    card_category TEXT,
    customer_tenure INTEGER DEFAULT 0 CHECK (customer_tenure >= 0)
);
```

```sql
-- 5. Create Fact_Transactions table
CREATE TABLE Fact_Transactions (
    transaction_id SERIAL PRIMARY KEY,
    customer_id INTEGER NOT NULL REFERENCES Dim_Customer(customer_id) ON DELETE
CASCADE,
    date_id INTEGER NOT NULL REFERENCES Dim_Date(date_id) ON DELETE CASCADE,
    transaction_amount NUMERIC(10,2) NOT NULL CHECK (transaction_amount >= 0),
    transaction_count INTEGER NOT NULL DEFAULT 1 CHECK (transaction_count >= 0)
);
```

Defined Measures:

Measure Name	Formula / Calculation	Description / Business Meaning
Total Transaction Amount	SUM(transaction_amount)	Total amount of digital payment transactions across all customers.
Total Transaction Count	SUM(transaction_count)	Total number of digital payment transactions.
Average Transaction Amount	AVG(transaction_amount)	Average value per transaction; shows typical transaction size.
Average Transactions per Customer	AVG(transaction_count)	Average number of transactions per customer; shows customer engagement.
Active Customer Count	COUNT([customer_id]) WHERE activity_status = 'Active'	Number of active customers using digital payments.
YOY Transaction Growth	(total transaction_amount this year - last year) / last	Year-over-year growth in transaction value.
MOM Transaction Growth	(total transaction_amount this month - last month) / last month	Month-over-month growth in transaction value.
Average Customer Tenure	AVG(customer_tenure_months)	Average duration (in months) customers have been active; indicates loyalty.
High-Value Customers	COUNT([customer_id]) WHERE transaction_amount >	Number of customers making high-value payments (VIP segment).
Low Activity Customers	COUNT([customer_id]) WHERE transaction_count < 5	Customers with very few transactions; potential churn risk.
Top Income Segment Revenue	SUM(transaction_amount) WHERE income_category = '$80k+'	Revenue contributed by the highest income segment; shows premium value.
Card Category Share	DIVIDE(SUM(transaction_amount), SUM(Card_Category_SUM(transaction_amount)))	Percentage of revenue per card type (Gold, Silver, etc.); highlights card performance
Gender-Based Transaction Split	DIVIDE(SUM(transaction_amount), SUM(transaction_amount)) by gender, SUM(transaction_amount_amount)	Shows which gender contributes more to digital payments; useful for targeting.
Average Transaction per Tenure Month	SUM(transaction_amount) / SUM(customer_tenure_months)	Average revenue generated per month of customer tenure; measures loyalty efficiency.
Dependents vs Transaction	SUM(transaction_amount) / AVG(dependents + 1)	Insight into how family size correlates with transaction volume.

2.3 Data Visualization

